AIDS-1 Unique Questions

Module 1: Introduction to AI & Intelligent Agents

#	Question	Appeared In
1	What is rational agent? Explain with diagram.	Dec 23
2	Explain learning agents with example.	Dec 24
3	Explain utility based agent architecture with diagram.	May 24
4	Explain goal based agents with example.	May 25
5	Describe PEAS and write PEAS representations for Medical diagnosis system.	Dec 23
6	State PEAS of automated taxi driver.	May 23
7	Explain PEAS descriptors. Provide PEAS descriptors for Part Picking robot.	May 25
8	What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.	Dec 24, May 25
9	Differentiate between data scientists, big data professionals and data analysts.	Dec 24, May 25
10	Differentiate between Forward and Backward chaining.	May 23
11	Explain working of Forward and Backward Chaining algorithms.	May 24
12	What are the different ways of knowledge representation?	May 23

Module 2: Problem Solving & Search Strategies

#	Question	Appeared In
1	Explain uniform cost search and best first search in detail with examples and compare.	Dec 23
2	Compare different search techniques based on their time complexities.	May 23

#	Question	Appeared In
3	State A *algorithm and explain with example how A* searching algorithm helps in finding the goal with optimal path.	May 23
4	Compare Informed and Uninformed Search algorithms. Explain working of A* algorithm with example.	May 25
5	Describe any four uninformed search strategies. Compare them with time complexity, space complexity, optimality and completeness.	May 24
6	Formulate the problem for 8 queens.	May 24
7	Explain state space based problem formulation. Formulate 8 Puzzle and N Queens problem.	May 25
8	Consider 5x5 grid with S(start), G(goal), #(obstacle), .(free). Represent as state space search problem. Apply Hill Climbing Search. Does the algorithm stuck in Local Minima?	May 24
9	Explain the working of the Hill climbing algorithm. What are issues in Hill climbing?	May 25
10	Can 1 liter water be measured using 10 liter and 4 liter jug? Justify.	May 23
11	Can min-max be used for team games? Draw sample trees for 2 and 3 teams.	May 23
12	Short note: Alpha Beta Pruning.	Dec 24
13	Short note: A* algorithm.	Dec 24
14	Short note: Min Max Algorithm.	May 25

Module 3: Knowledge Representation, Logic & Inference

#	Question	Appeared In
1	What is Unification? Give example.	Dec 23
2	What is Skolemization? Explain Skolem constant and Skolem function.	Dec 23
3	What is the difference between unification and Skolemization?	May 24
4	The law says it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is a criminal!	Dec 23
5	Marcus problem: Was Marcus loyal to Caesar? Solve using resolution.	May 23

#	Question	Appeared In
6	What are the rules of conversion from predicate to CNF? Explain each rule with proper example.	May 23
7	Short note: CNF.	Dec 24
8	Short note: Skolemization.	Dec 24
9	Short note: Rules in Propositional Logic.	Dec 24
10	Short note: Forward chaining based proofs.	May 25
11	What are heuristic functions? Where are they used?	Dec 23

Module 4: Exploratory Data Analysis (EDA) & Statistics

#	Question	Appeared In
1	Explain what role is played by Correlation and Covariance in EDA?	Dec 23
2	What do you mean by covariance and correlation? Explain the range of coefficients. Calculate COV and CORRCOV for given data. How do you interpret these values?	Dec 23, Dec 24
3	Compare barplot and histogram.	Dec 24
4	What is a histogram? Can we perform univariate graphical analysis using histogram?	May 23
5	Compare box-plot and scatter-plot.	May 24
6	What are the different univariate plots in EDA? Explain them in detail.	Dec 24, May 24, May 25
7	Explain various measures of the central tendencies of a statistical distribution.	May 23
8	With respect to Quantitative data analysis explain: i) Measure of central tendencies ii) Measure of spread iii) Skewness and Kurtosis.	May 23
9	Write a detailed note on Hypothesis Testing. What are type I and type II errors?	Dec 23
10	Compare Z-Test, T-Test and ANOVA in detail.	Dec 24, May 24
11	Perform one way ANOVA F Test on given house price prediction data and comment on whether the mean house price predicted by models A, B, C are same with level of significance 0.05.	May 23

#	Question	Appeared In
12	Perform t-test on given tea vs coffee customer preference data. Clearly mention Null and Alternate hypothesis. Comment whether t-test fails or succeeds in rejecting Null Hypothesis.	May 24
13	Consider given data. Draw scatter plot. Comment on the relationship. Find Coefficient of correlation. Does this statistic confirm your observation?	May 25
14	Short note: ANOVA.	May 25
15	Short note: Non graphical EDA.	May 25
16	Short note: Box Plot.	May 25

Module 5: Machine Learning Basics & Supervised Learning

#	Question	Appeared In
1	Write comparison between Business Intelligence and Data Science.	Dec 23
2	Write in detail issues in machine learning.	Dec 23, Dec 24, May 24, May 25
3	Elaborate in detail the steps in developing a Machine Learning application with architectural diagram.	Dec 23, Dec 24, May 25
4	Describe architecture of ML application. Explain with a diagram.	May 24
5	Define and state effects of overfitting and underfitting.	Dec 24, May 24, May 25
6	Define overfitting and underfitting in detail with a diagram.	May 24
7	What are the different types of Machine Learning algorithms? Give example of each category.	May 23
8	Explain SVM in detail.	Dec 23
9	What is SVM? Explain its significance in ML and compare it with logistic regression.	Dec 24
10	Write note on following supervised learning techniques: a) SVM b) ID3.	May 24
11	What is linear regression? Explain its significance in ML and compare it with logistic regression.	Dec 24, May 24

#	Question	Appeared In
12	Compare Linear Regression vs Logistic Regression with suitable diagrams and formulas.	May 23
13	In detail, explain steps in the Data Science Project.	May 23
14	What do you mean by data analytics? What are the different types of data analytics?	May 24
15	Explain various stages in the Data analytics Lifecycle.	Dec 23

Module 6: Advanced & Miscellaneous Topics

#	Question	Appeared In
1	What are the different planning techniques? Explain with example.	Dec 23
2	Short note: CNF.	Dec 24
3	Short note: Alpha Beta Pruning.	Dec 24
4	Short note: Skolemization.	Dec 24
5	Short note: Rules in Propositional Logic.	Dec 24
6	Short note: A* algorithm.	Dec 24
7	Short note: ANOVA.	May 25
8	Short note: Min Max Algorithm.	May 25
9	Short note: Non graphical EDA.	May 25
10	Short note: Forward chaining based proofs.	May 25
11	Short note: Box Plot.	May 25
12	What are the different issues in ML algorithms? (Repeated across papers – included in Module 5 as #2)	-